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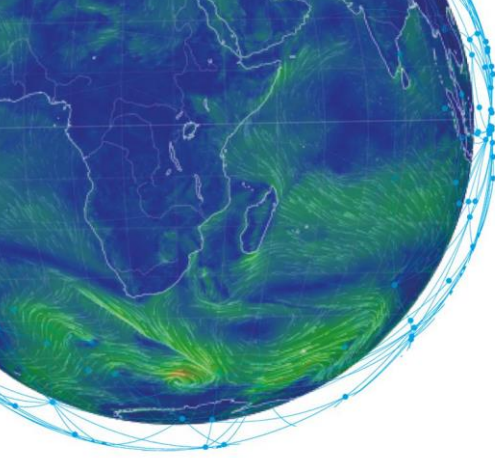
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Machine Learning in Weather & Climate

TRANSCRIPT

Why should we consider machine learning for weather and climate modelling?

Expert: Peter Dueben



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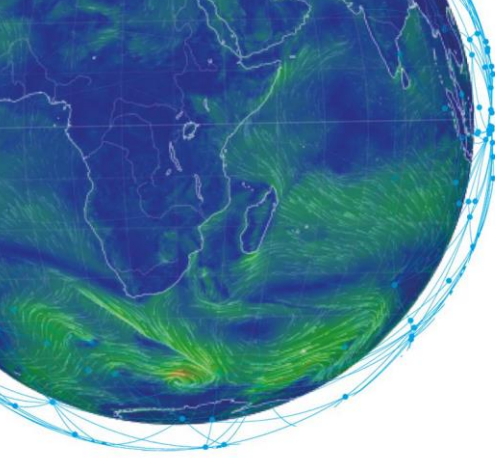
Introduction

Hello and welcome. My name is Lisa Burke and in this video we will learn more about how conventional weather and climate models are working and how they can be used to make predictions into the future. In this video we will have the pleasure to meet Peter Dueben who will tell us about the concept and state-of-the-art of those models. Peter is working at ECMWF and his section is developing global forecast systems that are used for predictions of the Earth system from days to centuries. Peter, the floor is yours.

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The Earth system is very complex with many components interacting with each other in a non-trivial way, including the atmosphere, the ocean, the land surface and sea ice, snow and ice on land, clouds and atmospheric chemistry. All those components need to be described in Earth system models which are used for numerical predictions of weather and climate. As a result, Earth system models are very complex and it requires an interdisciplinary effort and team to develop and maintain them, including Meteorologists, Physicists, Computing Scientists, Mathematicians and many more. Most of the physics that needs to be represented is well known, but some of the features, and in particular the interactions between the components, are still hard to describe by physical theory and even harder to describe in a numerical model.

Furthermore, the atmosphere and ocean show turbulent and chaotic behaviour. This results in an exponential increase of any errors or inconsistencies of the initial conditions even if the numerical model would be perfect. The numerical simulation of turbulent flow is already difficult for small problems such as a cup of coffee. It is extremely difficult for a system as big as the Earth.



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As a matter of fact, the Earth is so large that our global weather forecast models are typically working with around 10 km resolution and climate models are even coarser, despite the use of very large supercomputers to run those models. This resolution is not sufficient to resolve important processes such as individual clouds or the interactions between local topography and atmospheric motion.

In summary, the Earth system is not an easy target for conventional modelling.

On the other hand, a huge amount of data is available about the global Earth system, including both observations from many different sources, as well as model output data. At the European Centre for Medium-Range Weather Forecasts, we have an amount of data about the Earth system that is equivalent to hundreds of thousands of computer hard drives. This data holds a mesmerising amount of information about the Earth system and how the different components of the Earth system interact. Furthermore, the amount of data has been growing exponentially during the last couple of decades and will likely continue to do so in the future. Given the sheer size of the data, it is not easy to extract all relevant information and to relate the different datasets to each other.

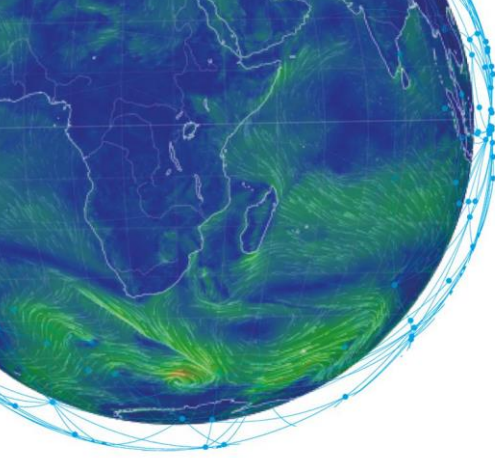
Here, machine learning can help. Machine learning allows scientists to extract information from very large amounts of data. It provides the capability to learn mappings between different datasets with no need for a comprehensive understanding of the physical theory. Such a tool is very useful in the context of Earth system modelling and can be applied in many ways.

Machine learning tools can be used to improve our understanding of the Earth system, for example when fusing information from different datasets such as measurements of surface temperature from a satellite, and surface temperature as represented within an Earth system model.

Machine learning can also be used for so-called unsupervised learning – for example to understand the causality of the interactions between different components of the Earth system. It is also possible to make components of Earth system models cheaper when running them on modern high-performance computers, to allow for a reinvestment of the savings and to push the models to higher resolution. This can, for example, be done when emulating model components with deep learning, or when using machine learning to compress model output data to be able to save more data at the same cost.

If tools are directly trained from observations, machine learning can also help to improve the models. This is, for example, possible when learning model errors or estimating uncertainties in predictions.

Finally, machine learning is a great tool for interdisciplinary research projects. You can take your favourite dataset from your favourite research area – for example a dataset from health applications or from the energy production of renewable energy power plants -- and learn how



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to relate the data to data from weather and climate models. For this, no knowledge of the explicit interactions or of the underlying theory will be required, which is often not available.

There are therefore many different applications for machine learning tools explored for across the workflow of weather and climate modelling, including the processing of observations, the process of data assimilation which is merging information of the model state and observations to generate initial conditions - and the post-processing of model output to make predictions usable by the general public via product generation.

It is now up to scientists in Meteorology, Oceanography, Biology, Numerics, applied mathematics and all related scientific areas in Earth system modelling, to explore how machine learning can be used to obtain the best predictions and to enhance our physical understanding. This journey has started around 5 years ago and is an ongoing process. This is not only true due to the complexity of the system under investigation and the amount of potential application areas, but also due to the breath-taking pace of developments in machine learning. Weather and climate models are already representing the Earth system at a breath-taking level of detail running on some of the fastest supercomputers of the world and progressing billions of variables in global simulations. I cannot wait to see the full potential of Earth system modelling unfold when those tools are combined intelligently with the power of modern machine learning.

Summary

This is very exciting: we are combining machine learning – as a tool which is extremely versatile and can basically be used wherever a large amount of data is available – with the scientific domain of weather and climate science, which is offering access to an unbelievable amount of data and tackling a system that is as complex as it gets on our planet. I can see that the study of machine learning in this context will keep us busy for a while.

Many thanks Peter for this great overview about the role of machine learning for weather and climate modelling. If you have any question or would like to discuss this subject in further detail, please use our forum on Moodle. It will be a pleasure to hear you!